

Assignment 3

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In [1]: import numpy as np

# Define input data and target values
input_data = np.array([[1,1,1,0,1,1,1], # 0
                       [0,0,1,0,0,1,0], # 1
                       [1,0,1,1,1,0,1], # 2
                       [1,0,1,1,0,1,1], # 3
                       [0,1,1,1,0,1,0], # 4
                       [1,1,0,1,0,1,1], # 5
                       [1,1,0,1,1,1,1], # 6
                       [1,0,1,0,0,1,0], # 7
                       [1,1,1,1,1,1,1], # 8
                       [1,1,1,1,0,1,1]]) # 9

target_values = np.array([[1,0], # Even
                           [0,1], # Odd
                           [1,0], # Even
                           [0,1], # Odd
                           [1,0], # Even
                           [0,1], # Odd
                           [1,0], # Even
                           [0,1], # Odd
                           [1,0], # Even
                           [0,1]]) # Odd
```

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In [2]: # Define the Perceptron Neural Network class
class PerceptronNN:
    def __init__(self, num_inputs, num_outputs):
        self.weights = np.zeros((num_inputs, num_outputs))
        self.bias = np.zeros(num_outputs)

    def activation(self, x):
        return 1 if x >= 0 else 0

    def feedforward(self, inputs):
        net = np.dot(inputs, self.weights) + self.bias
        output = np.array([self.activation(x) for x in net])
        return output

    def train(self, input_data, target_values, learning_rate, epochs):
        for epoch in range(epochs):
            errors = 0
            for i, input in enumerate(input_data):
                target = target_values[i]
                output = self.feedforward(input)
                error = target - output
                self.weights += learning_rate * np.outer(input, error)
                self.bias += learning_rate * error
            errors += np.abs(error).sum()
```

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In [3]: # Train the Perceptron Neural Network
perceptron_nn = PerceptronNN(num_inputs=7, num_outputs=2)
perceptron_nn.train(input_data, target_values, learning_rate=0.1, epochs=100)

# Test the Perceptron Neural Network on some sample inputs
input_test = np.array([[1,1,0,1,0,1,1], # 5
                       [1,1,1,0,1,1,1], # 0
                       [1,1,1,1,1,1,1]]) # 8

for input in input_test:
    output = perceptron_nn.feedforward(input)
    if output[0] == 1:
        print("Even")
    else:
        print("Odd")
```

Odd
Even
Even